

Affine Cipher

A third monoalphabetic cryptosystem.

The key consists of a pair $(\alpha, \beta) \in \mathbb{Z}_{26} \times \mathbb{Z}_{26}$ such that $\gcd(\alpha, 26) = 1$.

Encrypt a letter x using the following *affine function*:

$$x \mapsto (\alpha x + \beta) \pmod{26}.$$

Decrypt as follows:

$$x \mapsto \alpha^{-1}(x - \beta) \pmod{26}.$$

Note: Of course, you can define the affine cipher over any $\mathbb{Z}_m$.

Questions about the Affine Cipher

How does the affine cipher relate to the shift cipher and the substitution cipher?

How many possible keys are there?

Is the affine cipher secure?

What is α^{-1} ?

Why do we have the condition: $\gcd(\alpha, 26) = 1$?

Are the encryption and decryption functions efficiently computable?

How would you cryptanalyze the affine cipher?

Cryptanalysis of the Affine Cipher

We will look at a ciphertext only attack.

- Brute-force (like the shift cipher, but now we have 312 possible keys).
- Frequency analysis (like the substitution cipher).
- Two letters of plaintext and two corresponding letters of ciphertext usually suffice to find the key. This approach is taken in a known plaintext, chosen plaintext, or chosen ciphertext attack, but can also be used in a ciphertext only attack. In that case, the pairs are guessed from the frequency analysis.

Affine Cipher

Recall:

The key consists of a pair $(\alpha, \beta) \in \mathbb{Z}_{26} \times \mathbb{Z}_{26}$ such that $\gcd(\alpha, 26) = 1$.

Encrypt a letter x using the following *affine function*:

$$x \mapsto (\alpha x + \beta) \mod 26.$$

Decrypt as follows:

$$x \mapsto \alpha^{-1}(x - \beta) \mod 26.$$

For example, suppose we guess that 4 is encrypted to 17 and 19 is encrypted to 3. Show that this guess must be incorrect. See Section 3.3 for general background on congruences.

If we guess that 4 is encrypted to 17 and 19 is encrypted to 10, the key must be $(3, 5)$. You should then decrypt the ciphertext with $(3, 5)$ to see if this guess was correct.

Even if you have only one plaintext letter and the corresponding letter of ciphertext, this will reduce the number of possible keys to 12. You can then brute-force.

Some Number Theory Related to the Affine Cipher (see Chapter 3)

There are 12 elements a of $\mathbb{Z}_{26}$ such that $\gcd(a, 26) = 1$.

The number of integers in $\mathbb{Z}_m$ that are relatively prime to m is often denoted by $\phi(m)$. ϕ is often called the *Euler phi function*.

So, $\phi(26) = 12$.

If p is prime, what is $\phi(p)$?

In general, if $m = \prod_{i=1}^n p_i^{e_i}$, where the p_i 's are distinct primes, and $e_i > 0$ for $1 \leq i \leq n$, then

$$\phi(m) = \prod_{i=1}^n (p_i^{e_i} - p_i^{e_i-1}).$$

What is the number of keys in the affine cipher over $\mathbb{Z}_m$?

Computing $a^{-1} \pmod{m}$

Method 1: For every $a' \in Z_m$, compute aa' . If $aa' \equiv 1 \pmod{m}$, then $a' \equiv a^{-1} \pmod{m}$.

Disadvantage: Really slow! And we will need an efficient algorithm later.

We can do much better, using the *extended Euclidean Algorithm*.

Some Number Theory Related to the Affine Cipher (see Chapter 3)

The extended euclidean algorithm given a, m gives us the following equation:

$$as + mt = \gcd(a, m)$$

Assuming $\gcd(a, m) = 1$, we then have the following.

$$\begin{aligned} as + mt &= 1 \\ as &= 1 - mt \\ as &\equiv 1 \pmod{m} \\ a^{-1} &\equiv s \pmod{m} \end{aligned}$$

We will first look at the Euclidean algorithm, which can be used to compute the greatest common divisor.

Euclidean Algorithm

Euclidean Algorithm (a, b) .

// a and b are positive integers

$r_0 := a$

$r_1 := b$

$m := 1$

while $r_m \neq 0$ do

$q_m := \lfloor \frac{r_{m-1}}{r_m} \rfloor$

$r_{m+1} := r_{m-1} - q_m r_m$

$m := m + 1$

$m := m - 1$

// $r_m = \gcd(a, b)$

It is not hard to show that

$\gcd(r_0, r_1) = \gcd(r_1, r_2) = \dots = \gcd(r_{m-1}, r_m) = r_m$.

Extended Euclidean Algorithm

Since a has a multiplicative inverse $\pmod{b}$ if and only if $\gcd(a, b) = 1$, we can use the Euclidean algorithm to determine whether or not a has a multiplicative inverse $\pmod{b}$.

To *compute* the inverse, we can use the extended Euclidean algorithm, which, given positive integers a and b , computes integers r , s , and t such that $r = \gcd(a, b)$ and $sa + tb = r$.

How does this help you to compute $a^{-1} \pmod{b}$?

Well, if $\gcd(a, b) = 1$, then $sa + tb = 1$. Taking this equation $\pmod{b}$, we obtain $(sa + tb) \pmod{b} = 1 \pmod{b} = 1$. Since $sa + tb \equiv sa \pmod{b}$, it follows that $sa \pmod{b} = 1$, and thus $a^{-1} \pmod{b} = s \pmod{b}$.

Extended Euclidean Algorithm

Extended Euclidean Algorithm (a, b) .

// a and b are positive integers

$r_0 := a; r_1 := b$

$t_0 := 0; t_1 := 1$

$s_0 := 1; s_1 := 0$

$m := 1$

while $r_m \neq 0$ do

$q_m := \lfloor \frac{r_{m-1}}{r_m} \rfloor$

$r_{m+1} := r_{m-1} - q_m r_m$

$t_{m+1} := t_{m-1} - q_m t_m$

$s_{m+1} := s_{m-1} - q_m s_m$

$m := m + 1$

$m := m - 1$

// $r_m = \gcd(a, b)$ and $s_m a + t_m b = r_m$.

It is not hard to prove (by induction on j) that for all $0 \leq j \leq m$,

Solving $ax \equiv c \pmod{m}$

See page 72; useful, for example, when cryptanalyzing the affine cipher.

- If $\gcd(a, m) = 1$, then the solution is

$$x \equiv a^{-1}c \pmod{m}.$$

- If $\gcd(a, m) = d > 1$ and d does not divide c , then there is no solution.
- If $\gcd(a, m) = d > 1$ and $d|c$, then let x_0 be the solution of the congruence

$$(a/d)x \equiv (c/d) \pmod{m/d}.$$

The solutions of the original congruence are

$$x \equiv x_0 + i(m/d) \pmod{m},$$

for $i = 0, \dots, d - 1$.

Exercise 1.21 (a) from Stinson

EMGLOSUDCGDNCUSWYSFHNSFCYKDPUMLWGYICOXYSIPJCK
QPKUGKMGOLICGINCGACKSNISACYKZSCKXECJCKSHYSXCG
OIDPKZCNKSHICGIWYGKKGKGOLDSILKGOIUSIGLEDSPWZU
GFZCCNDGYYSFUSZCNXEOJNCGYEOWEUPXEZGACGNFGLKNS
ACIGOIYCKXCJUCIUZCFZCCNDGYYSFEUEKUZCSOCFZCCNC
IACZEJNCSEHFZEJZEGMXCYHCJUMGKUCY

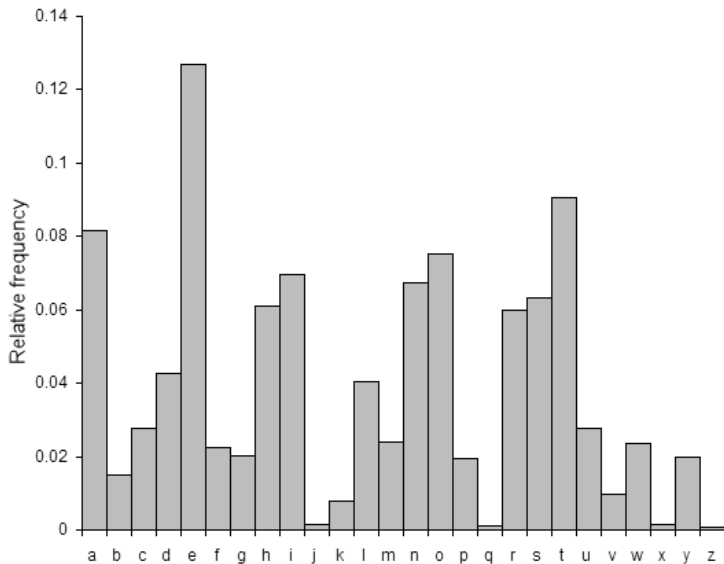
Hint: F decrypts to w.

Frequency Analysis

SINGLE CHARACTER FREQUENCIES

A	5	N	13
B	0	O	10
C	37	P	6
D	8	Q	1
E	12	R	0
F	9	S	20
G	24	T	0
H	5	U	14
I	15	V	0
J	7	W	5
K	18	X	7
L	7	Y	15
M	5	Z	13

English Letter Frequencies



Digrams and Trigrams

All digrams that occur more than once:

- 7: CG, ZC
- 5: CK, CN, GO, NC, YS
- 4: CY, FZ, GK, GY, SF
- 3: CC, CI, CJ, GL, IC KG, KS, KU, MG, OI, SH, SI, US, XC, XE, ZE
- 2: CF, CS, DG, DP, DS, EJ, EO, EU, GA, GI, IG, IU, JC, JN, JU, KX, KZ, LK, ND, NS, OL, PK, SA, UC, UG, UM, UZ, WY, YK, YY

All trigrams that occur more than once.

- 3: CCN, FZC, GOI, YSF, ZCC
- 2: CFZ, CGI, CJU, CKS, CKX, CND, CYK, DGY, GAC, GOL, GYY ICG ICK INC KGO KSH NCG NDG SAC UZC

Digram and Trigram Stats

The 30 most common digram are (in decreasing order):

TH, HE, IN, ER, AN, RE, ED, ON, ES, ST, EN, AT, TO, NT, HA, ND, OU, EA, NG, AS, OR, TI, IS, ET, IT, AR, TE, SE, HI, and OF.

The most common 12 trigrams are:

THE, ING, AND, HER, ERE, ENT, THA, NTH, WAS, ETH, FOR, and DTH.